

Partition-Tolerant Federation Across Emulated ISP Islands

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Abstract—An Internet shutdown is rarely a clean power switch. Transit to the outside world can be withdrawn while domestic networks keep carrying packets among themselves, leaving reachable fragments that recent measurement work calls connectivity islands. We ask a deliberately narrow question: when some domestic IP paths survive, can independently operated forum servers keep accepting signed posts and keep reconciling them across whatever graph remains? Our prototype combines four mechanisms that are individually familiar and, to our knowledge, have not been composed for this operating regime: canonical protobuf envelopes admitted as the exact bytes their author signed, operator receipts bound to those bytes, caller-initiated synchronisation that survives asymmetric reachability, and an explicitly trusted dual-homed relay between islands. On a four-node testbed whose shared exchange segment is withdrawn mid-run, a dependent certificate, community, and post crossed the severed boundary in 0.5, 1.2, and 2.1 seconds—within a factor of two of the same chain measured while the exchange was intact. An eight-node chain delivered 200 of 200 envelopes seven hops with a 4.125s median and a 6.115s 99th percentile, and per-hop cost tracked a queueing model to within 89ms across a fourfold change in the drain interval. TypeScript, Rust, and Python implementations agreed on all 16 canonical vectors. A signature-invalid flood produced no database writes, though a rate limiter shed 90% of it before the pipeline was reached, which we report rather than headline. Against a median gzipped ActivityPub activity our envelopes are $4.4\text{--}6.9\times$ smaller, at the cost of Fediverse interoperability and JSON-LD extensibility. These results establish partition tolerance under surviving IP paths on a single-host testbed. They establish nothing about total blackouts, real inter-provider deployment, or resourced denial of service.

Index Terms—federated systems, Internet shutdowns, network partitions, signed objects, store-and-forward

I. INTRODUCTION

Treating an Internet shutdown as a binary event obscures what operators actually do. Disruptions arrive as throttling, destination blocking, protocol filtering, or the withdrawal of international transit, and they frequently leave domestic reachability partly intact [1], [2]. Baltra *et al.* formalised the resulting structures, naming networks severed from the Internet core *islands* and persistent partial reachability *peninsulas* [3]. Their contribution is measurement: they detect and name the condition. Ours is a system that has to keep working inside it. The question we pose is correspondingly narrow. If domestic IP paths survive a transit cut, can independent forum servers continue to admit signed posts and reconcile them across the surviving graph?

Adjacent systems answer adjacent questions. Circumvention tools such as Telex and Snowflake route a user to an external service through cooperating infrastructure or ephemeral prox-

ies [4], [5]; both presume that some path out of the country exists, which is precisely what a transit cut removes. Device-to-device systems—Anix, Rangzen, Moby, Twimight—abandon the packet network entirely and carry messages on Bluetooth or Wi-Fi Direct encounters [6]–[9], which is the right approach for a total blackout and costs latency, bandwidth, and battery to take. Neither family specifies how independently operated *domestic servers* should validate, store, and relay public forum objects while partial IP connectivity persists. That gap is what this paper occupies.

Our design rests on a single discipline: an author signs one exact byte sequence, and every server that later handles those bytes verifies them unchanged. No relay may normalise, repair, or re-encode an object on its way through. Acceptance produces a receipt signed over the same bytes, so an operator’s acknowledgement is bound to what was actually admitted rather than to an identifier the operator chose. Synchronisation is always initiated by the caller, so a node with no reachable inbound address still both publishes and recovers. When two islands cannot reach each other at all, an explicitly trusted dual-homed server validates, stores, and relays between them, running the same admission code as every other node rather than forwarding packets.

We claim four contributions:

- a byte-preserving admission path applied identically to locally submitted and federated objects, which removes the failure mode in which a relay repairs a malformed object into a verifying one;
- caller-initiated delivery, streaming, and backfill, giving bidirectional federation to nodes that can open connections but not accept them;
- a quota-bounded relay policy that revalidates every object it forwards and excludes the peer it arrived from, so trust governs rate rather than validity; and
- measurements from a testbed in which the inter-island path is withdrawn mid-run, together with cross-language byte-agreement vectors, a hop-scaling chain, a signature-invalid flood, and a compression-aware ActivityPub size comparison.

We claim *partition tolerance under surviving IP paths* and nothing broader. The design cannot manufacture a route through a total communication blackout, cannot conceal a visible relay from the network operator hosting it, and cannot compel an operator to accept a valid post.

II. BACKGROUND AND RESEARCH GAP

Federated and signed systems. ActivityPub delivers social activities to server inboxes [10]; Matrix and the AT Protocol federate rooms and replicate repositories [11], [12]; Nostr and Secure Scuttlebutt replicate author-signed content [13], [14]. Each supplies a mechanism we reuse. None assumes that the server graph itself may fragment: reachable peer addresses remain a background assumption, and no member of the family combines exact-byte admission, operator receipts, outbound-only recovery, and an explicit two-island relay policy in the arrangement evaluated here.

Blackout communication. Anix, Rangzen, Moby, and Twimight move data through physical proximity [6]–[9]; Dolphin smuggles data through a cellular voice channel when packet service is denied [15]. These designs trade latency, throughput, and energy for independence from any domestic server path. Ours makes the opposite trade and inherits the mirror-image weakness: it is fast while servers remain mutually reachable and it stops entirely when every IP path is gone. The two families are complements, not competitors, and we make no throughput or latency comparison against them because the workloads are not commensurable.

Accountability. Certificate Transparency established append-only logs, signed tree heads, and inclusion proofs as a way of making certain misbehaviour visible after the fact [16], [17]; PeerReview and CoSi pursue stronger notions through witnesses and cosigning [18], [19]. Our receipts are a deliberately weaker instrument in that lineage: a receipt proves that one operator acknowledged one specific byte string at one moment, not that the operator accepts all valid objects, forwards them promptly, or refrains from silently declining the next one.

The gap sits between normally routed federation, which assumes the server graph holds, and device-to-device networks, which assume it has vanished. We assume instead that servers stay powered and at least one domestic path survives, and the objective under that assumption is to admit an authored object once, preserve its signed bytes without alteration, and let every reachable replica reach the same verdict on its validity.

III. SYSTEM DESIGN

A. Architecture

A local client and a remote peer converge on the same ingress path; nothing about the transport that carried an object changes how it is judged. Admission inspects the submitted bytes before any persistent write occurs. After the transaction commits, the server returns a receipt and enqueues federation work in a durable outbox, which a scheduler drains grouped by destination, source uplink, and traffic class. A bridge is an ordinary server running that same stack, dual-homed onto two networks. It is not a transparent packet forwarder, and the distinction matters: everything it relays, it has first independently validated and stored under its own key.

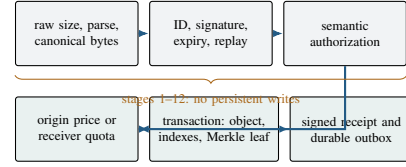


Fig. 1. Admission pipeline. State changes begin only after the cheap byte and validity checks have all succeeded.

B. Signed objects and admission

An envelope carries a domain tag, an object type, a payload, a public key, a nonce, and an expiry. Its identifier is the SHA-256 digest of a specified canonical byte string, and an Ed25519 signature covers that same string [20]. Because protobuf does not guarantee a unique serialisation [21], we constrain a profile on top of it: ascending field order, minimal variants, no unknown fields retained, and—the operative rule—a decode followed by a re-encode that must reproduce the received bytes exactly. An object that fails that comparison is rejected rather than repaired.

Strict-decode-then-verify-what-arrived is long-established practice in DER, JWS, and CT’s leaf format; what it buys here is specific to relaying. A federation node that decoded a peer’s bytes and re-encoded them before verification would silently normalise a malformed object into a well-formed one and then find that it verifies. Under our rule the receiver never normalises attacker-controlled bytes before checking the signature, so no relay can launder an object into validity.

Fig. 1 summarises the admission path. Stages 1 through 12 cover size, parsing, canonicity, domain separation, identifier derivation, signature, expiry, replay, and authorisation, and none of them writes to persistent storage—an invalid flood therefore cannot amplify into write load, a property we measure in §V. Later stages charge origin pricing or a receiver quota, commit the raw object together with its derived indexes in a single transaction, append its hash to a Merkle log, and only then enqueue federation. A uniqueness constraint in the database, rather than a read-then-write check, makes exact retries idempotent under concurrency.

On acceptance the server signs a receipt binding the object identifier, its leaf position, the resulting tree head, and an inclusion proof, so the acknowledgement commits to the bytes admitted rather than to a label the operator is free to change afterwards. Peers additionally exchange signed tree heads. Two heads of equal size with differing roots are an unambiguous conflict; a larger head is recorded as pending until a consistency proof can be fetched. We are explicit about the reach of this rule. It is stale-head hygiene that eliminates a class of self-inflicted false positives—and it does catch outright tree shrinkage—but an adversary who rewrites history while keeping the tree non-decreasing passes both clauses undetected, and equivocation between two peers cannot be caught by any purely local comparison. Detecting either requires consistency proofs and cross-observer exchange, which the partition we are designing for is precisely what obstructs.

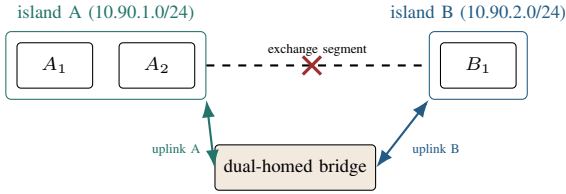


Fig. 2. Route-cut topology. Four application nodes span two isolated network segments joined by a shared exchange, which the experiment withdraws mid-run. The bridge is an application server, not an IP forwarder.

C. Partition-aware federation

Three remote procedure calls, all initiated by the caller, cover ordinary delivery and recovery. `DELIVER` pushes a batch, `STREAMACTIVITIES` resumes from a durable cursor, and `BACKFILL` requests objects a node knows it is missing. Because none of the three requires the calling node to accept an inbound connection, a server behind carrier-grade network address translation participates fully in both directions over connections it opens itself. This is reachability asymmetry, not discovery: at least one peer address must already be known.

The scheduler drains independent per-peer queues concurrently under a bounded lease, with deadlines, exponential backoff, and idempotency keys. That structure exists because its absence was a real defect, described in §V. One caveat is worth stating in the design rather than buried in limitations: the deadline bounds how long a drain pass waits on a peer, not how long the underlying transport keeps trying, because the sender takes no cancellation token. A stuck call can therefore hold resources past its deadline even though it no longer blocks anyone.

Fig. 2 shows the route-cut topology. Two ordinary nodes sit in island A and one in island B; a shared exchange segment connects them, standing in for an Internet exchange point. The bridge holds one address on each island network. Once the exchange is withdrawn, both islands must already have configured trust in the bridge—trust that governs the rate and priority a peer’s traffic receives, never whether its objects are valid. The bridge validates and stores every object it handles, bounds volume per uplink pair and per traffic class, and never relays an object back to the peer it arrived from.

Four implementation rules carry most of the replication safety. The transport hands ingress an opaque byte string and has no ability to decode and re-encode it. Every peer, the bridge included, passes the complete validity check. Database uniqueness, not application logic, absorbs the duplicate that inevitably arrives when a live stream and a backfill both carry the same object. Finally, fan-out excludes the originating peer, which avoids a redundant round trip that content deduplication would eventually terminate but not for free.

IV. METHODOLOGY

We evaluate five questions: whether independent encoders agree on the canonical form, whether objects cross an enforced route cut, how delivery latency scales with hop count, whether

malformed requests produce storage writes, and how the representation compares in size with ActivityPub. All experiments run on one physical host; §VI treats the consequences of that at length.

Correctness tests. A shared corpus of 16 positive and adversarial byte vectors is executed independently by the TypeScript, Rust, and Python implementations, each of which must agree byte-for-byte on encoding and identifier derivation and must reject every negative vector. A further 85 focused tests cover ingress (29), the outbox (3), federation (31), and simulated ISP behaviour (22). These are regression tests written by the system’s authors, not an independent security audit.

Route-cut experiment. Four application nodes, two databases, and two caches are distributed across three isolated virtual network segments as in Fig. 2. Containers supply reproducible layer-2 isolation and let the harness withdraw the only configured inter-island path mid-run; nothing in the design depends on that packaging. The gate reads kernel TCP state from `/proc/net/tcp` inside the bridge to confirm which local source address each established federation connection actually left from, withdraws the exchange segment, verifies that direct probes between islands now fail and that island B does not already hold the content, submits a dependent certificate, community, and post, and polls for each exact content identifier. Nineteen assertions cover topology, source binding, relay behaviour, quota reservation, and uplink failover. Reported latencies include the application’s own polling interval and are therefore upper bounds.

Hop-scaling chain. Eight nodes are peered into a line, each with its own database, so that a node i hops from the origin is reachable by exactly one path of exactly that length. Hop count is the independent variable, which a mesh would destroy, since in a mesh every node is one hop away and only fan-out width varies. Latency is computed by subtracting the `received_at_ms` value each node writes inside the same transaction as its projection, so no polling observer sits in the measurement path; this is sound only because all nodes share the host clock, which a real deployment would have to replace with NTP discipline and an error bar. We inject 200 valid envelopes at hop 0 at a paced rate and record arrival at hop 7. The eight nodes share a single database process with one database each, which introduces contention and inflates the reported figures—the conservative direction in which to be wrong.

Signature-invalid flood. Random bytes are refused within microseconds at the parse stage and measure nothing interesting. The tool therefore lifts a genuine envelope from the node’s own log and mutates a single byte inside the signed region, which invalidates the signature and changes the content identifier, so every request is novel and none can be short-circuited by deduplication. We issue 3,000 such requests at concurrency 16 and record completion rate, rejection stage, the fraction reaching signature verification, and—via the database’s collection-level profiler—the resulting writes. The ordinary rate limiter stays enabled, so this is not a saturation

benchmark.

Encoding comparison. We collected 80 real Create/Note activities from the public outboxes of two stock ActivityPub deployments. Because a collection hoists @context to the collection while an activity POSTed to a remote inbox carries its own, measuring the bare outbox item would understate delivery size; the tool refetches a standalone object from the same host and re-attaches exactly those bytes. We compare raw and gzipped activity bytes against equivalent signed envelopes, subtracting authored UTF-8 text so the metric reflects what a protocol designer controls rather than how much a user typed. The comparison is deliberately generous to ActivityPub, excluding HTTP framing, TLS, per-follower fan-out, shared dictionaries, and—most significantly—authentication, since our bytes carry a 64-byte signature verifiable at rest while HTTP Signatures are transport level and vanish with the request.

V. RESULTS

A. Correctness and convergence

Every implementations agreed on all the 16 canonical vectors and rejected all the negative ones as specified. The additional tests also checked invalid signatures, incorrect byte formats, replay attacks, missing permissions, duplicate messages, wrong signing keys, invalid proofs, outdated tree heads, and conflicting logs. Sixteen vectors cannot cover every protobuf’s edge cases. However, they show that those three independently developed implementations follow exactly the same byte-level rules. This is essential for a relay network to work reliably.

The route-cut test passes all 19 checks. The TCP connection check also confirmed that the federation connections used both configured addresses, 10.90.1.30 and 10.90.2.30. But this could only be verified with real network sockets, not an in-process test. When direct communication between the islands was disabled, Island B still received the certificate after 0.5 s, the community after 1.2 s, and the post after 2.1 s. This follows their dependency order through the bridge. For comparison, with the direct connection still available, the same items took 1.5 s, 1.5 s, and 1.9 s. So the bridge worked correctly and was still within about twice the speed of the direct connection. These are results from single test runs, and not statistical averages.

The 2.1 s result was achieved through a specific fix and was not just simply observed. An earlier version passed 18 of 19 tests, but the one successful crossing took 407.6 s, with all outbox records still showing zero delivery attempts. The problem was that peers were contacted sequentially with no time limit. If one peer was unreachable, it blocked delivery to all the others. This includes the bridge that was supposed to bypass the network failure. We were able to fix this by delivering to peers concurrently and adding a time limit to each delivery. This reduced the crossing time from 407.6 s to 2.1 s, which is a 195× improvement.

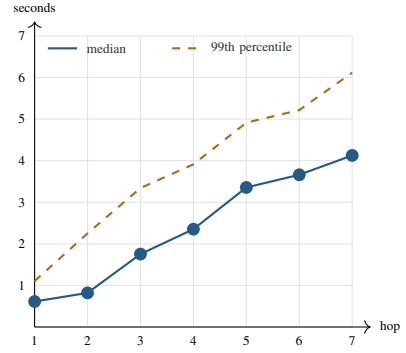


Fig. 3. Measured per-hop latency across the eight-node chain at a 1,000 ms drain interval, from in-transaction timestamps. All 200 envelopes reached hop 7.

B. Latency scaling and admission cost

All 200 envelopes successfully traveled through the eight-node chain to hop 7. The median End-to-end latency was 4.125 s and the 99th percentile 6.115 s. The per hop results which are in Fig. 3 are more useful than the final numbers. Most of the delay comes from the outbox drain interval, not the protocol itself, which if predicted roughly then it’s $drain/2$ plus a fixed term; With a 1,000 ms drain interval, each hop took about 589 ms, including roughly an 89 ms fixed protocol work. With a 250 ms interval, each hop took about 196 ms, implying 71 ms of fixed work. Since this fixed cost stayed around 71–89 ms even when the drain interval changed, it actually represents the protocol’s actual per-hop cost. This includes verifying, projecting, appending, and re-queuing the message. The remaining delay is mainly a scheduling choice that operators can adjust based on bandwidth and battery needs. However, these results should not be treated as Internet performance predictions because the test used a shared database process and did not include real wide-area network delays.

The invalid-signature flood was rejected at 730 requests per second. This was all done with no writes to objects, indexes, logs, or the outbox. The test also confirmed that the profiler was actually running as it recorded six read operations during that test. Of the 3,000 total requests, about 2,700 were blocked by the per-peer rate limiter. So, only around 9% reached signature verification, and then the expensive rejection process, which is the canonical parsing followed by Ed25519 verification. This ran about 66 requests per seconds. The final conclusion is a bit limited but clear enough. Invalid traffic was stopped before causing any writes, with the rate limiter handling most of the attack first. This test does not necessarily show how the system would handle floods from many different peers, valid messages that should cause writes, or long-term parser and database pressure.

Table I compares two encoding formats and their costs. Both tested hosts produced the same 927-byte @context declaration, which implies this is a fixed cost of the standard ActivityPub format, not something specific to our deployment. The declaration alone is more than four times larger than our

entire signed forum post. Compression removes much of the apparent size advantage, so we use compressed ActivityPub numbers for a fair comparison. Gzipped ActivityPub messages were 786–1,400 B with a median of 1,074 B. Our raw signed envelopes were only 155 B, 220 B, and 243 B making ActivityPub about 4.4–6.9 $\times$ larger. Our envelopes do not benefit from compression at all as the 64-byte Ed25519 signature is incompressible and gzip makes all three slightly *larger*. So the honest comparison is gzipped ActivityPub against our raw bytes, not the roughly 20 $\times$ ratio suggested by comparing uncompressed sizes. However, size is the only area where our format is better. It cannot connect to the existing Fediverse, requires generated codecs and unfamiliar tools, does not allow unknown fields, and therefore requires versioned schemas for future changes.

VI. DISCUSSION AND LIMITATIONS

Testbed validity. Every experiment ran on one physical host using virtual network segments, and the resulting evidence ceiling is the paper’s principal weakness. All nodes share a kernel, clock, CPU, and storage subsystem; the shared clock is what makes cross-node timestamp subtraction valid at all, and it is exactly what a real deployment does not provide. Virtual segments, static addressing, and local name resolution are far simpler than physical routers, BGP reconvergence, carrier-grade NAT, DNS failure, variable MTUs, and radio access, and attaching a dual-homed bridge to two virtual networks is trivial where a real one needs separate links, routing policy, credentials, power, and an operator willing to be identified. The route-cut experiment therefore demonstrates a change in software and routing state, not geographic latency, independent failure domains, or provider behaviour.

Access and subscriber isolation. The design assumes nodes within an island can reach one another or a shared bridge. Wi-Fi client isolation at an access point, or subscriber isolation within an ISP, can block customer-to-customer traffic while leaving every customer able to reach the provider gateway. Where that isolation holds on every surviving path, nodes can neither dial each other nor reach a bridge, federation fails entirely, and only isolated local service remains. The prototype has no non-IP fallback for this case.

Network adversary. A network operator can enumerate stable endpoints, block addresses or SNI values, fingerprint the RPC traffic, correlate timing, isolate a node, or coerce a bridge operator. Source-address binding selects an uplink; it conceals nothing. One shutdown model defeats us outright and deserves naming rather than omission: protocol whitelisting, in which domestic routing stays intact but deep packet inspection admits only sanctioned protocols. Our federation traffic is a distinctive non-standard protocol on a distinctive port, so under whitelisting it dies on the first day regardless of how sound its signatures are. Carrying federation over ordinary HTTPS, rotating endpoints, and adopting pluggable transports are the obvious responses, and none of them is implemented or evaluated here.

System and application limits. Signatures authenticate keys, not truthfulness or safety, and a legitimate key can publish abuse or an expensive flood. Because admission cost is charged at the origin, an origin operator controls its own secrets and can mint unlimited free proofs for its own users; anti-abuse strength is therefore bounded by the policy of the weakest origin in a peer set, and per-peer quotas are the only backstop. Key theft, metadata exposure, moderation at scale, revocation recovery, storage exhaustion, and coordinated Sybil behaviour all remain open. Transport deadlines bound waiting rather than cancelling work. Consistency proofs for growing trees are specified but not implemented. Most fundamentally, a receipt evidences only what was acknowledged: an operator who declines admission outright leaves no receipt and hence no evidence, so the cheapest form of censorship—silence—is the one this construction cannot document.

Evaluation limits. The route-cut figures come from single preserved runs under one configuration, and the encoding sample is 80 activities of a single type from two deployments. We have no multi-host, wide-area, multi-day, or real-shutdown trials, and did not preserve host load, repeated-run distributions, or confidence intervals. Resource figures were taken on x86 hardware and say nothing about the single-board computers a community deployment would realistically use. Comparisons against proximity-based and voice-channel systems concern different workloads and are not a ranking.

Ethics. All tests used synthetic content on isolated private networks. A real deployment could expose bridge operators and users to legal or physical risk, since a visible relay is by construction legible to the network that hosts it, and should not be attempted in a hostile setting without local threat analysis, informed consent, data minimisation, and community review.

VII. CONCLUSION

Server federation can survive a route cut when domestic IP paths remain. Canonical signed objects, a single admission path for local and federated traffic, caller-initiated recovery, and an explicitly trusted validating relay carried a dependent post across a severed exchange in 2.1 s, within a factor of two of the intact path, and the defect that stood between 407.6 s and that figure was visible only once three peers were deployed with one of them unreachable. Cross-language byte agreement and a hop-scaling chain whose per-hop cost tracks a queueing model to within 89 ms support the mechanism from two further directions. The compact representation is 4.4–6.9 $\times$ smaller than gzipped ActivityPub and forfeits interoperability, extensibility, and mature tooling to get there. The work we consider most urgent is multi-host and wide-area evaluation, consistency proofs for growing logs, cancellation-aware transport, floods of validly signed traffic, redundant independent bridges, transport camouflage against protocol whitelisting, and a non-IP fallback for subscriber isolation and total blackouts.

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TABLE I

MEASURED REPRESENTATION SIZE AND DESIGN TRADE-OFFS AGAINST ACTIVITYPUB. THE BYTE SAMPLE IS 80 REAL CREATE/NOTE ACTIVITIES FROM TWO STOCK DEPLOYMENTS. RATIOS COMPARE GZIPPED ACTIVITYPUB AGAINST RAW SIGNED ENVELOPES, WHICH DO NOT COMPRESS.

Dimension	This work	ActivityPub comparison	Cost or boundary
Measured bytes	Raw signed envelopes: check-in 155 B, forum post 220 B, broadcast 243 B; gzip enlarges all three	Delivered JSON-LD median 4,216 B, gzipped median 1,074 B (range 786–1,400 B); @context alone a fixed 927 B	Ratio 4.4–6.9×; representation only—HTTP, TLS, fan-out, shared dictionaries, and AP authentication all excluded in AP’s favour
Interoperability	Custom protobuf domains and RPCs	Standard actor, object, and inbox model	No Fediverse compatibility and no existing application ecosystem
Evolution	Unknown fields rejected; versioned profiles required	Flexible JSON-LD vocabulary and open extension	Harder rolling upgrades and more cross-language test surface
Operations	Generated codecs and canonical-byte tooling required	Text inspectable with ordinary web tools	Higher debugging and tooling burden on operators
Integrity profile	One exact byte string per object, content ID, author signature, and receipt	Authentication depends on deployment profile; HTTP Signatures do not survive the request	Unambiguous byte identity, but a new security profile to maintain and audit

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